Gayatri Vidya Parishad College of Engineering Autonomous

Computer Science and Engineering | Course Code: CS201

Data Structures and Algorithms - Laboratory Manual & Practical Guide

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Syllabus Focus: CS201 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Asymptotic Analysis & Linear Structures

" Syllabus Scope: Big-O, Omega, Theta asymptotic bounds, Master Theorem, Amortized analysis, Stacks, Queu

Unit 2: Balanced Search Trees & Heaps

" Syllabus Scope: Binary Search Trees, AVL rotations, Red-Black tree coloring rules, Binary Heaps, Priori

CS201: Data Structures and Algorithms - Advanced Topics & Assessment

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Graph Traversal & Minimum Spanning Trees

" Graph representations, BFS, DFS, Topological sorting, Prim and Kruskal Minimum Spanning Tree algorithms, Dis

Unit 4: Shortest Path Algorithms

" Dijkstra single-source shortest path with priority queues, Bellman-Ford algorithm for negative weights, Floy

Unit 5: Dynamic Programming & Greedy Design

" Optimal substructure, 0/1 Knapsack, Longest Common Subsequence LCS, Matrix Chain Multiplication, Huffman cod

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Implementation of Doubly Linked List with Insertion, Deletion, and Reversal Operations in C++

Lab 2: Construction of AVL Tree with Automatic Left, Right, Left-Right, and Right-Left Rotations

Lab 3: Graph Traversal Implementation using BFS and DFS with Cycle Detection

Lab 4: Implementation of Dijkstra Shortest Path Algorithm using Binary Min-Heap Priority Queue

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Thomas H. Cormen, Charles E. Leiserson - Introduction to Algorithms MIT Press 3rd Ed

* Mark Allen Weiss - Data Structures and Algorithm Analysis in C++ Pearson 4th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students